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Cryptography Lab: Exploring Encryption with CyberChef

Objective:

- To understand basic encryption techniques using CyberChef and explore different cryptographic methods.
- Tools: CyberChef (https://gchq.github.io/CyberChef/)

Introduction to CyberChef:

Interface Overview:

- Input Window: On the left side, where you input your text or data.
- Recipe Window: In the middle, where you drag and drop operations to create your "recipe."
- Output Window: On the right side, where you see the result of your recipe.

Using Basic Operations:

- Select an Operation: Use the search bar at the top to find the operation you need (e.g., "Base64," "ROT13," "XOR").
- Drag and Drop: Click on the desired operation and drag it to the Recipe window.
- Configure Settings: If the operation requires any settings (e.g., key for encryption), you can adjust them in the Recipe window.
- Run the Recipe: Type or paste your data into the Input window, and CyberChef will automatically process it through your recipe. The output will appear in the Output window.

Name(id): ศุภณัฐ พลยมมา 2513160032

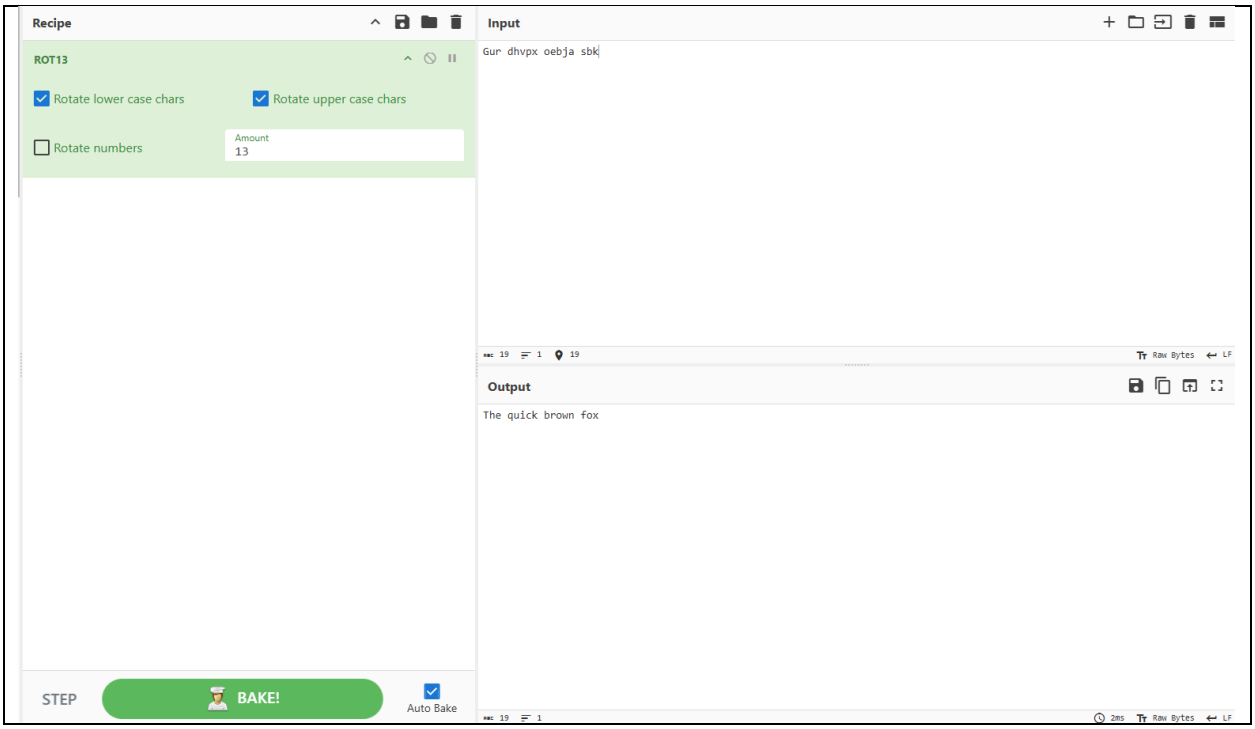
Exercise 1: Caesar Cipher

Task: Encrypt and decrypt a message using the Caesar Cipher.

Steps:

- Go to CyberChef and search for "ROT13" (a type of Caesar Cipher).
- Drag the ROT13 operation to the Recipe window.
- Type a message (e.g., "The quick brown fox") into the input window and click "Bake."
- Observe the encrypted output
- Reverse the process to decrypt the message.

Show your CyberChef Screen:



Name(id): ศุภณัฐ พลยมมา 2513160032

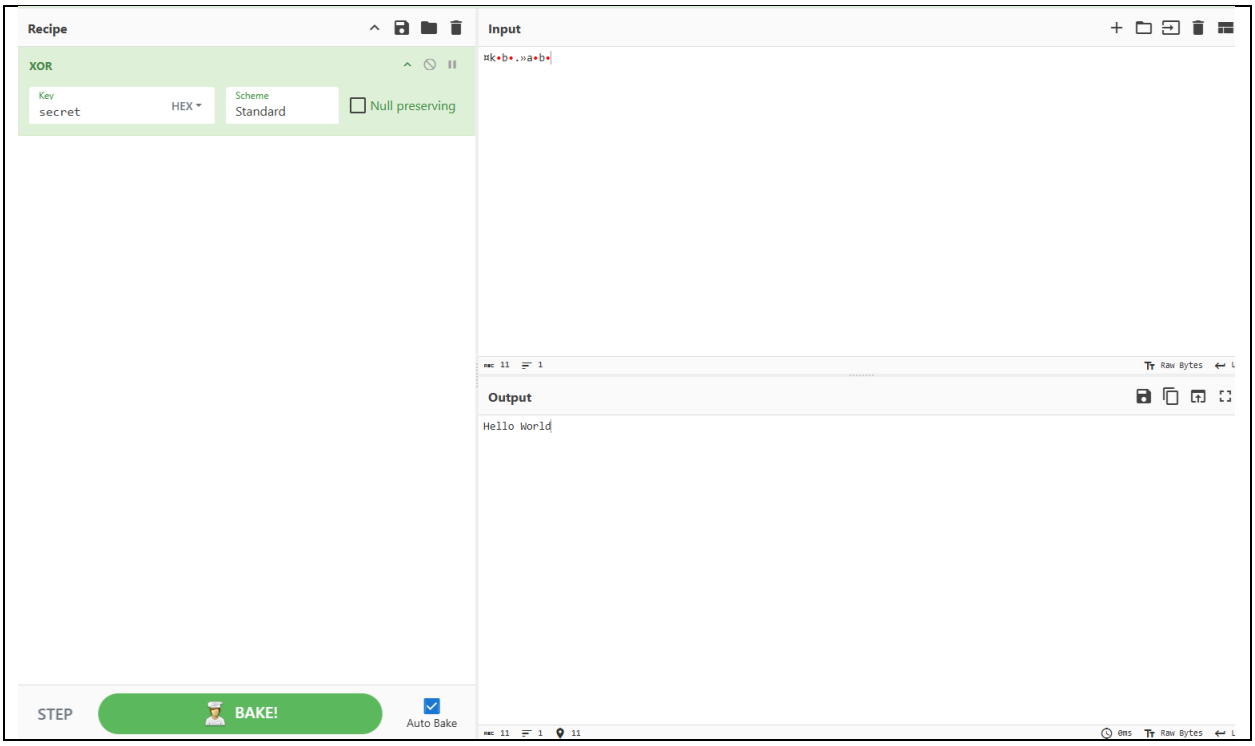
Exercise 2: XOR Encryption

Task: Encrypt and decrypt a message using XOR encryption.

Steps:

- Go to CyberChef and search for "XOR."
- Drag the XOR operation to the Recipe window.
- Enter a key (e.g., "secret") and a message (e.g., "Hello World").
- Click "Bake" to see the encrypted output.
- Use the same key to decrypt the message.

Show your CyberChef Screen:



Name(id): ศุภณัฐ พลยมมา 2513160032

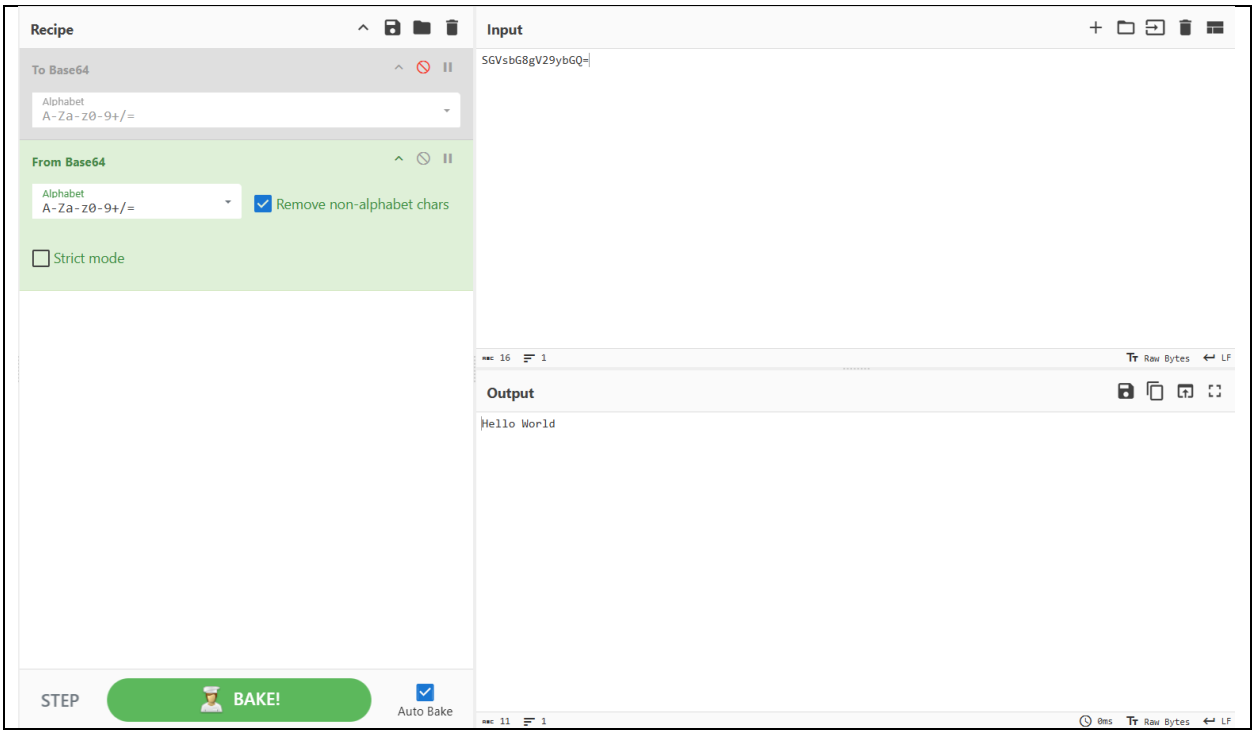
Exercise 3: Base64 Encoding

Task: Encode and decode a message using Base64.

Steps:

- Go to CyberChef and search for "Base64."
- Drag the Base64 operation to the Recipe window.
- Enter a message (e.g., "Hello World") and click "Bake."
- Observe the Base64 encoded output.
- Decode the Base64 encoded message back to its original form.

Show your CyberChef Screen:



Name(id): ศุภณัฐ พลยมมา 2513160032

Exercise 4: Advanced Encryption (AES)

Task: Decrypt a message using AES encryption.

Hint: What is Fork operation?

Input:

U6cA0twJSNzilOOdU/rtFA==

kT98GoiMu2KnBosB4AJWJmwmg66g8+OvBvSUopTzDn8=

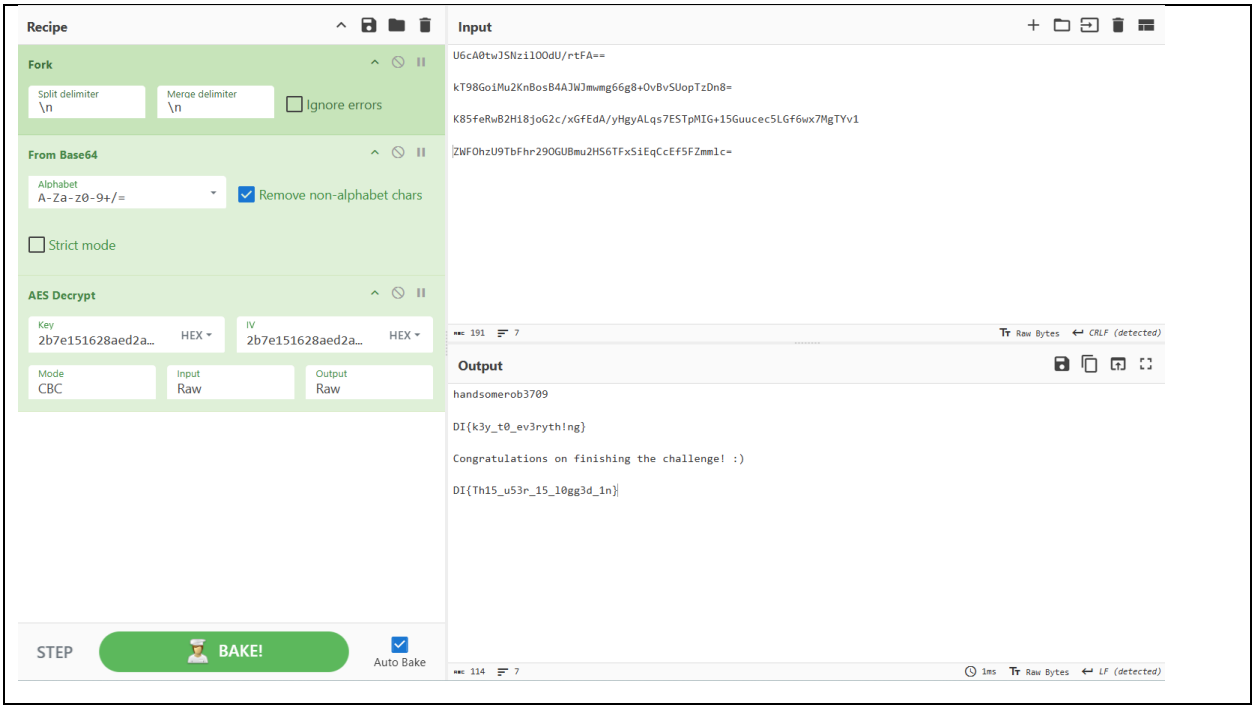
K85feRwB2Hi8joG2c/xGfEdA/yHgyALqs7ESTpMIG+15Guucec5LGf6wx7MgTYv1

ZWFOhzU9TbFhr29OGUBmu2HS6TFxSiEqCcEf5FZmmlc=

Key = 2b7e151628aed2a6abf7158809cf4f3c

IV = 2b7e151628aed2a6abf7158809cf4f3c

Show your CyberChef Screen:



Name(id): ศุภณัฐ พลยมมา 2513160032

Exercise 5: Public Key Cryptography with CyberChef

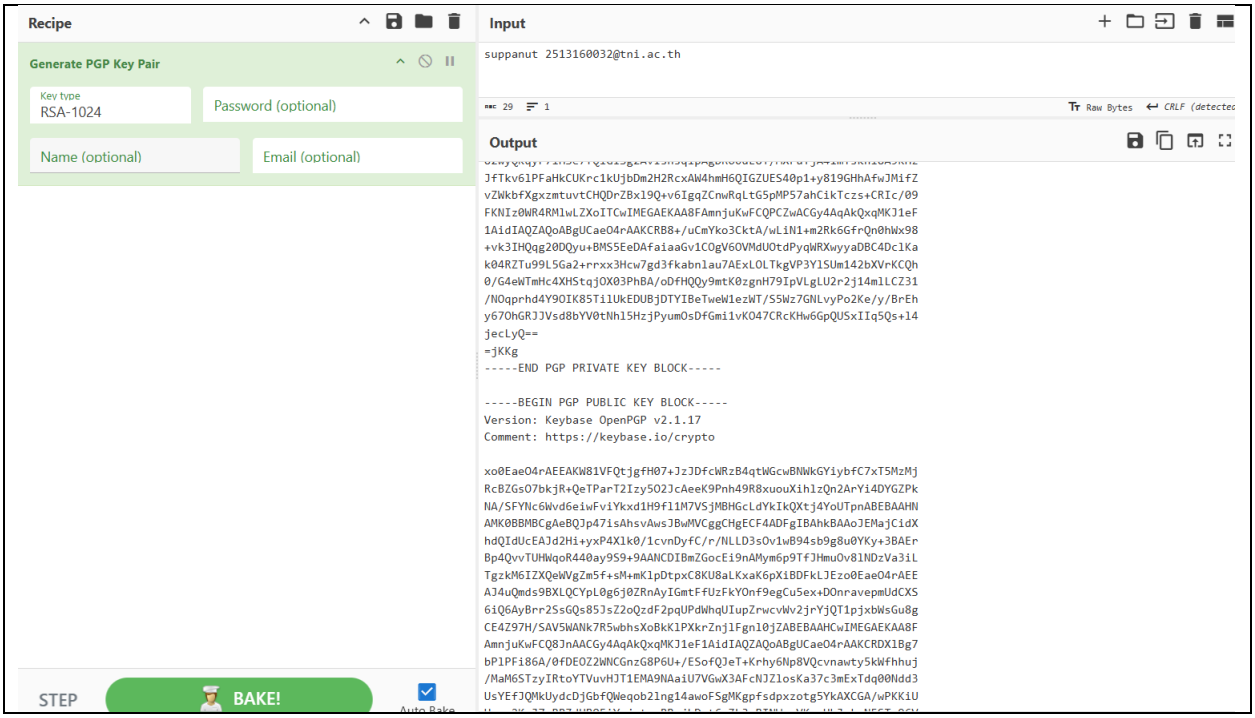
Generating PGP Key Pair:

Task: Generate a PGP key pair using CyberChef.

Steps:

- Go to CyberChef and search for "Generate PGP Key Pair."
- Drag the "Generate PGP Key Pair" operation to the Recipe window.
- Enter your name and email address, then click "Bake."
- Observe the generated public and private keys.

Show your Public and Private key:



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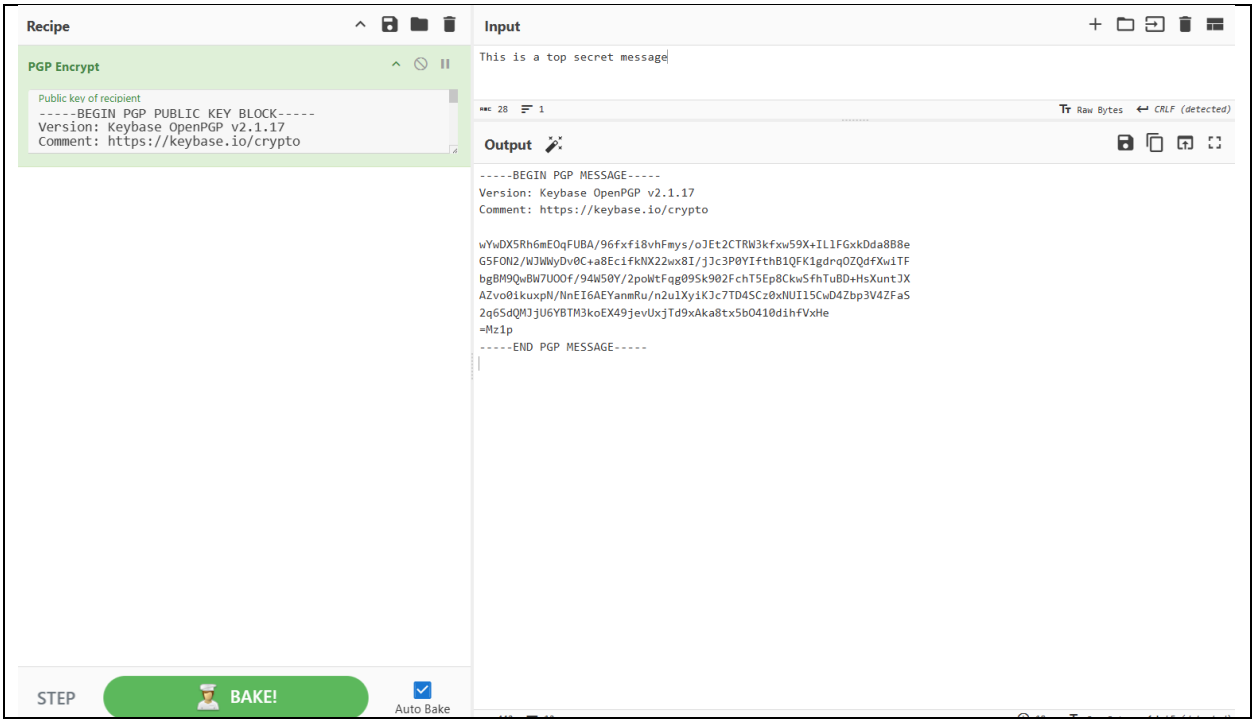
Task: Encrypt a message using the public key.

Steps:

- Go to CyberChef and search for "PGP Encrypt."
- Drag the "PGP Encrypt" operation to the Recipe window.
- Enter the public key and the message you want to encrypt.
- Click "Bake" to see the encrypted output.

Show your CyberChef Screen:



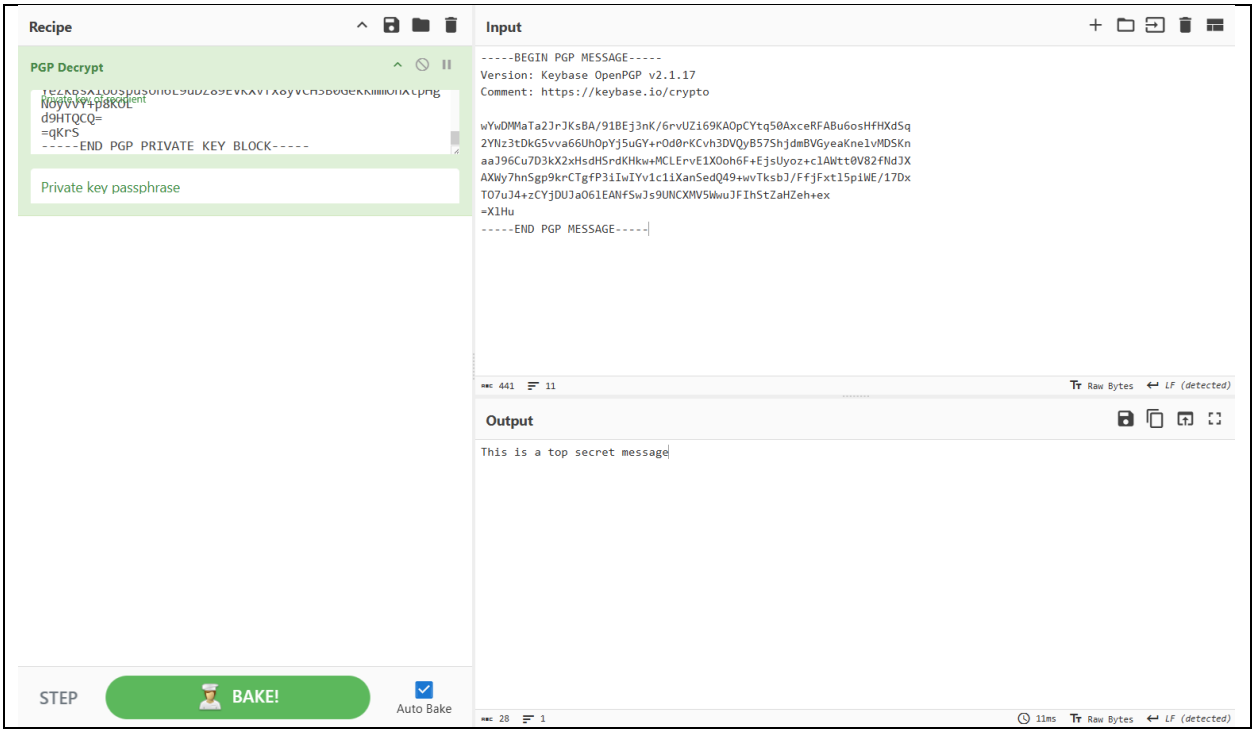
Name(id): ศุภณัฐ พลยมมา 2513160032

Task: Decrypt the encrypted message using the private key.

Steps:

- Go to CyberChef and search for "PGP Decrypt."
- Drag the "PGP Decrypt" operation to the Recipe window.
- Enter the private key and the encrypted message.
- Click "Bake" to see the decrypted output.

Show your CyberChef Screen:



Name(id): ศุภณัฐ พลยมมา 2513160032

Discussion Questions:

What are the advantages and disadvantages of each encryption method?

Ans: **ROT13:** ใช้ง่ายและเร็วที่สุด แต่ความปลอดภัยต่ำมาก เพราะไม่มีกุญแจล็อค เหมาะสำหรับซ่อนข้อความทั่วไปไม่ให้คนเผลออ่าน

AES: รวดเร็วและปลอดภัยสูงมากสำหรับข้อมูลจำนวนมาก แต่มีจุดอ่อนเรื่องความเสี่ยงหากกุญแจที่ใช้ร่วมกันหลุดไปถึงคนอื่น

PGP: ปลอดภัยสูงสุดและจัดการกุญแจได้ง่ายกว่าเพราะไม่ต้องส่งรหัสให้กัน แต่ขั้นตอนการใช้งานซับซ้อนและกินทรัพยากรเครื่องมากกว่าแบบอื่น

How does the complexity of the key affect the security of the encryption?

Ans: กุญแจที่ยิ่งซับซ้อนจะยิ่งเพิ่มจำนวนรูปแบบในการเดาให้มากขึ้นมหาศาล จนคอมพิวเตอร์ต้องใช้เวลาสุ่มรหัสนานหลายล้านปี ทำให้การเจาะข้อมูลทำไม่ได้ในทางปฏิบัติ

Why is it important to use strong encryption methods in real-world applications?

Ans: การเข้ารหัสที่แข็งแกร่งช่วยปกป้องข้อมูลส่วนตัวและการเงินจากการถูกขโมยหรือแก้ไข พร้อมทั้งสร้างความเชื่อมั่นว่าข้อมูลจะส่งถึงผู้รับที่ถูกต้องอย่างปลอดภัย